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Letter of Recommendation of *Hyeonbeen Lee*

Dear Ilju Foundation Committee,

I am pleased to offer my strongest recommendation for Hyeonbeen Lee for the Ilju Foundation Overseas Ph.D. Scholarship. I have closely supervised and mentored him for the past five years since 2021, first as a student in my courses, then as an undergraduate researcher, a Master's student, and most recently as a postgraduate researcher in my laboratory. This unusually long and continuous association has allowed me to evaluate him not only as an excellent student but as a researcher of exceptional promise.

I first came to know him in my undergraduate courses on computational dynamics and finite element methods, where he consistently ranked among the best students. What distinguished him was not only his academic performance but also the depth of his curiosity. He regularly went beyond the assigned material by verifying derivations computationally and turning course-related methods into more reproducible implementations. He did not stop at completing a given task, but tried to understand the structure of the problem and improve the quality of the work itself. That was the main reason I accepted him into my lab as an undergraduate researcher.

During his Master's studies, his growth as an independent researcher became clear. In our work on neural-network-based modeling of nonlinear dynamics, he identified a significant limitation in the existing framework for contact-rich systems and proposed a new multi-order learning architecture. This work led to his first-author publication in the *Journal of Computational Physics* and also received the Excellent Paper Award from the *Korean Society of Mechanical Engineers*. His excellence during the Master's program was further recognized through institutional financial support, including approximately USD 27,000 in graduate funding through Brain Korea 21 Plus and a selective Graduate Assistantship. These accomplishments reflected not only strong ability but also outstanding rigor and originality for a student at that stage.

He later extended this line of research to a more demanding real-world robotics problem involving learning-based force estimation for a hydraulic manipulator. What began as a modest estimation task in simulation gradually evolved into a highly ambitious project using real sensor data and sequential forecasting. Before graduating, he developed a decomposition-based probabilistic framework for predicting contact-rich wrench dynamics in vibration-intense environments, thereby improving both the performance of the model and the methodological level of our lab's machine learning research. This project also became the foundation of the work that he continued and matured further *after graduation*.

Rather than ending the project upon completing his Master's degree, he chose to carry it forward while pursuing his own career, even without continued research funding and formal supervision. During this postgraduate period, he improved the inference speed for real-time deployment, incorporated newer modeling methods, and developed the project into a more impactful contribution to the literature. This effort has led to his second first-author manuscript, which has recently been submitted to *IEEE/ASME Transactions on Mechatronics*. Through this period, I came to realize more clearly that he continues to pursue meaningful research with strong passion, self-motivation, and persistence, even in the absence of external support.

In summary, Hyeonbeen is one of the strongest students I have mentored. He has already demonstrated originality, rigor, and research ability through his publications and awards. What distinguishes him even more, however, is that he continued to advance his research after graduation through his own initiative, even without continued funding and supervision. In my view, this rare combination of proven ability, sustained self-motivation, and long-term commitment makes him exceptionally well-suited for doctoral study. He also stands out as the first student from my group to enter a Ph.D. program in the United States. For a foundation committed to supporting scholars of both ability and character, I believe he is an exceptionally worthy candidate. I therefore recommend him to the Ilju Foundation without reservation and with full confidence in his future as a doctoral researcher.

Sincerely,

Jin-Gyun Kim